

DIMITROVGRAD, Serbia

WMO: 133970

Lat: 43.017N

Lon: 22.750E

Elev: 1476

StdP: 13.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	15.5	5.8	7.8	13.5	10.5	9.9	16.9	16.4	30.0	14.5	30.5	4.8	110	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	91.6	67.9	88.4	67.1	85.2	66.1	70.7	85.4	69.1	83.4	67.7	81.3	6.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	100.9	76.9	64.3	95.4	74.9	62.9	90.7	73.1	35.5	85.2	34.1	83.7	32.9	81.4	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	12.9	11.6	DB	2.6	96.3	6.7	3.4	-2.2	98.8	-6.1	100.8	-9.9	102.7	-14.8	105.2
			WB	2.3	73.4	6.5	2.2	-2.4	75.1	-6.2	76.4	-9.9	77.6	-14.7	79.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	31.6	35.4	42.5	51.5	59.6	66.0	69.5	69.9	61.8	52.4	43.8	34.2	(6)
	DBStd	15.18	8.44	8.92	8.02	7.10	6.14	5.90	5.07	5.28	6.57	7.64	8.05	8.19	(7)
	HDD50	2078	571	411	253	63	6	0	0	0	2	66	214	492	(8)
	HDD65	5360	1036	828	697	405	184	58	17	17	136	392	635	955	(9)
	CDD50	2665	0	2	21	109	304	479	605	617	355	141	30	2	(10)
	CDD65	474	0	0	0	1	17	87	157	169	40	3	0	0	(11)
	CDH74	5775	0	0	1	58	319	1061	1788	1975	521	51	1	0	(12)
	CDH80	2122	0	0	0	9	56	348	705	839	160	5	0	0	(13)
(14) Wind	WSAvg	5.7	6.2	6.3	6.4	6.1	5.5	5.1	5.1	5.1	5.2	5.4	6.0	6.2	(14)
(15) Precipitation	PrecAvg	24.8	1.7	1.6	1.7	2.2	2.8	2.7	2.2	2.0	2.0	2.2	1.9	1.8	(15)
	PrecMax	38.3	4.2	3.1	3.5	5.7	6.2	5.6	4.9	6.9	6.1	5.1	5.0	4.3	(16)
	PrecMin	14.1	0.3	0.3	0.2	0.3	0.9	0.6	0.3	0.2	0.2	0.2	0.0	0.0	(17)
	PrecStd	5.6	1.1	0.8	0.9	1.1	1.2	1.3	1.2	1.7	1.4	1.4	1.1	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	64.6	71.7	80.2	85.1	91.6	96.2	96.2	90.6	79.2	69.4	58.3	(19)
		MCWB	46.8	49.8	53.5	59.2	63.9	69.1	69.6	67.8	65.0	61.2	56.6	50.5	(20)
	2%	DB	49.8	57.6	66.4	74.2	80.8	87.4	91.4	91.9	84.5	74.2	63.9	53.1	(21)
		MCWB	44.0	47.4	50.7	56.3	62.6	68.4	68.1	67.8	63.5	59.3	53.9	47.3	(22)
	5%	DB	46.1	53.1	61.7	69.6	77.5	84.1	87.6	88.6	80.1	70.1	59.5	49.0	(23)
		MCWB	41.8	45.1	48.8	54.0	61.2	67.0	67.3	66.7	62.1	57.9	51.7	44.9	(24)
	10%	DB	42.4	48.3	57.0	65.6	73.8	80.5	83.9	85.0	75.5	66.0	55.7	45.6	(25)
		MCWB	39.3	42.7	46.7	52.4	59.9	65.9	66.7	65.7	60.8	55.9	49.8	42.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	51.1	54.7	60.7	66.2	72.3	72.8	72.6	67.2	63.4	57.6	52.1	(27)
		MCDB	53.6	61.8	67.8	77.1	79.6	86.1	87.5	87.8	83.0	75.6	66.6	57.1	(28)
	2%	WB	44.7	48.1	51.6	57.6	64.1	70.0	70.6	70.3	64.9	61.0	54.7	47.9	(29)
		MCDB	49.2	55.9	63.3	70.6	76.9	83.9	85.5	85.4	79.6	71.3	62.3	52.4	(30)
	5%	WB	42.1	45.7	49.5	55.4	62.6	68.2	69.1	68.6	63.5	58.8	52.7	45.1	(31)
		MCDB	45.8	52.4	60.0	66.9	75.0	81.3	82.9	83.9	76.5	67.7	58.7	48.7	(32)
	10%	WB	39.4	43.0	47.3	53.6	60.8	66.4	67.6	66.9	61.9	56.8	50.3	42.6	(33)
		MCDB	42.2	48.0	56.2	63.4	72.0	78.6	80.7	81.3	73.1	64.4	55.5	45.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	16.2	19.4	21.4	22.5	23.5	24.9	26.0	22.7	20.1	16.0	12.4	(35)
		MCDBR	18.9	25.1	29.0	29.8	29.7	30.2	32.0	32.8	30.8	27.1	22.6	18.5	(36)
		MCWBR	13.1	15.3	15.7	14.2	13.5	13.1	12.4	11.9	12.6	14.0	13.3	12.7	(37)
	5% WB	MCDBR	17.4	22.9	26.1	25.3	26.6	27.4	27.4	27.9	26.2	23.7	19.9	16.6	(38)
		MCWBR	12.3	14.2	14.5	12.8	13.0	12.8	12.2	11.9	11.9	13.1	12.0	11.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.336	0.369	0.418	0.486	0.470	0.458	0.464	0.480	0.449	0.402	0.364	0.336	(40)
	taud		2.277	2.188	2.113	1.952	2.057	2.137	2.127	2.077	2.128	2.246	2.312	2.314	(41)
	Ebn at Noon		246	257	258	249	257	260	257	248	245	242	234	235	(42)
	Edh at Noon		28	36	44	56	52	48	48	49	43	34	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		483	752	1084	1407	1698	1938	1997	1794	1302	895	555	411	(44)
	RadStd		72	102	134	171	155	169	145	156	109	120	64	61	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+1.21	N/A	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	
Regional (39 neighbors)	+0.94	N/A	N/A	N/A	+0.67	N/A	N/A	-263	N/A	N/A	

Nomenclature: See separate page